Advanced Planning & Scheduling for AVEVA Manufacturing Execution System

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What is Advanced Planning & Scheduling

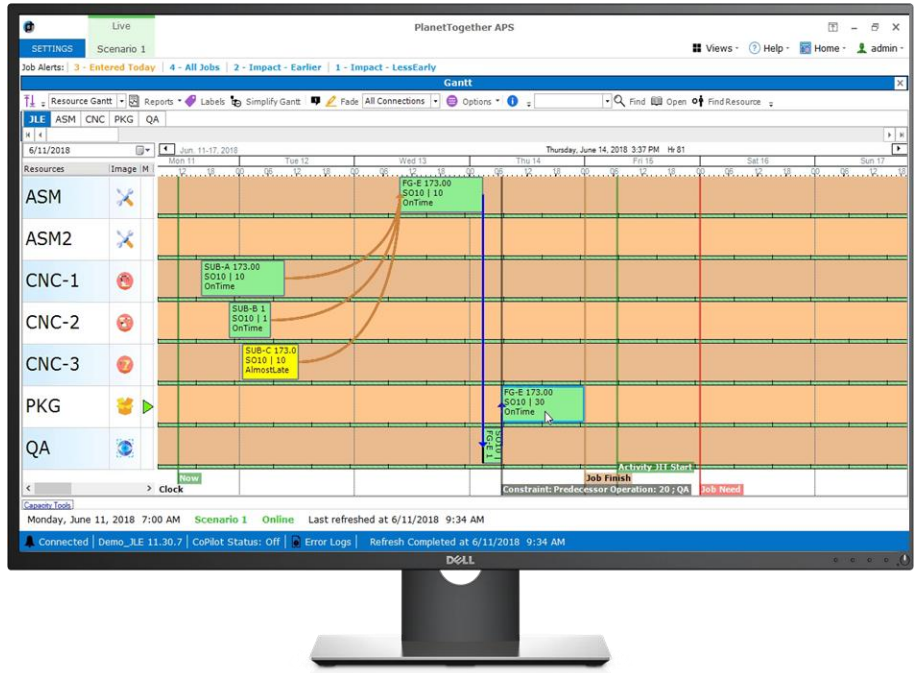
How can Manufacturers benefit?

- **Advanced Planning and Scheduling**: A production planning process by which raw materials and production capacity are optimally allocated to meet demand.

- **Challenge**: Traditional MRP systems plan Materials and Capacity separately, do not adjust for capacity or material constraints, and don't create detailed work sequences that take setup time into account -- leading to infeasible and inefficient plans.

- **Solution**: Advanced Planning and Scheduling from PlanetTogether simultaneously plans based on available materials, labor and machine capacity, and automates the optimization of sequence-dependent production schedules.

- **Result**: Optimized feasible plans, created in seconds



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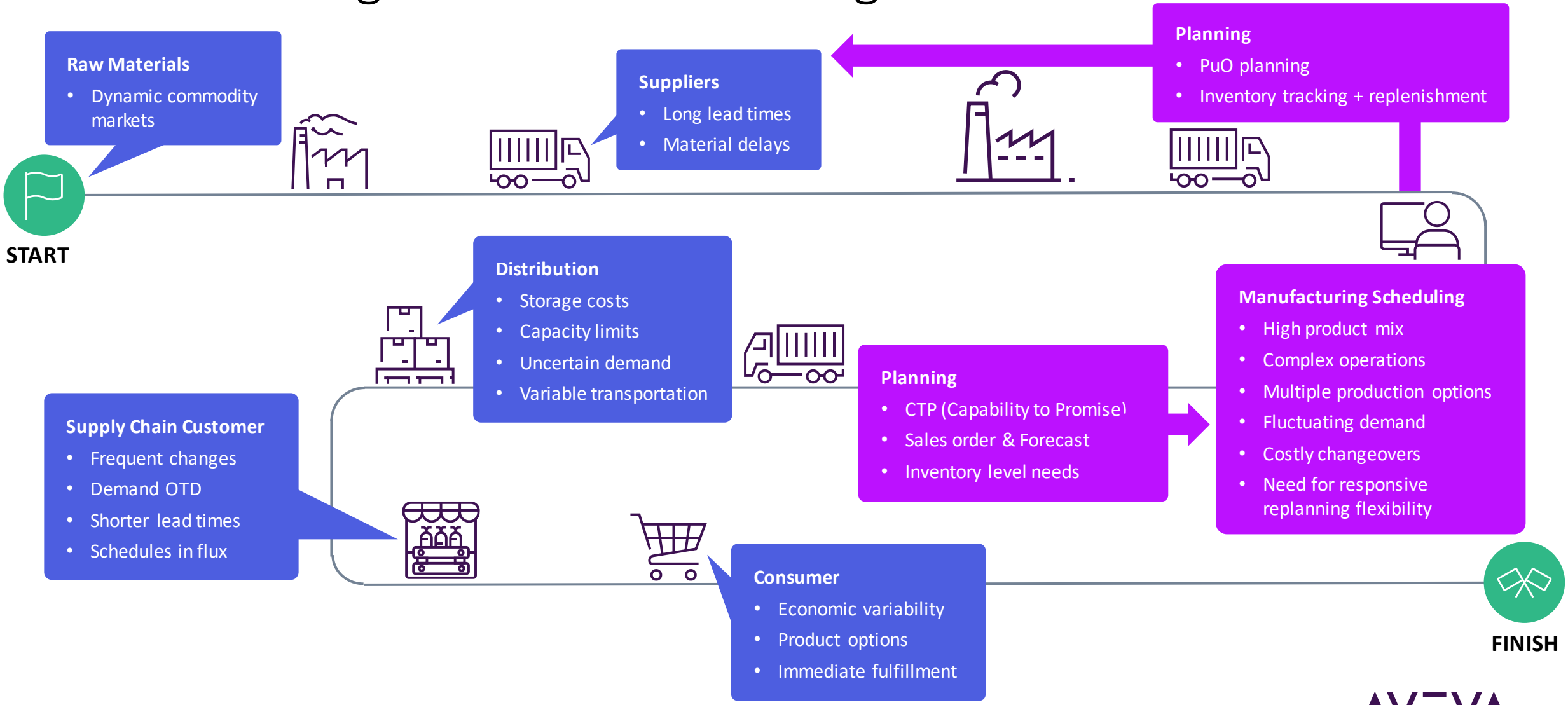
Manufacturing Value Chain Challenges



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Integrated Manufacturing Planning and Execution

At A Glance:

In todays markets manufacturing operations need to be dynamically optimized to ensure the maximum business returns that operational constraints allow. Within the plants, manufacturing operations must be continuously optimized to minimize losses and to deliver on time in full.

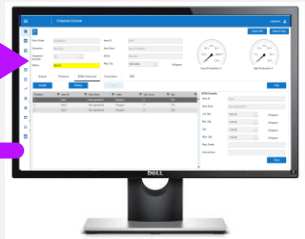
Plan

Create production plans that are feasible to meet business goals

Schedule

Continuously optimize plant schedules for throughput and on-time delivery

Execute



Execute on the plan to deliver on-time, in-full

- Integration of Advanced Planning and Scheduling software from PlanetTogether with AVEVA Manufacturing Execution System allows for feasible production plans based on actual plant conditions and a continuously optimized production schedule that balances business KPI's with operational constraints.
- The AVEVA Manufacturing Execution System aligns people, equipment and operational processes for reliable, sustainable & compliant execution of the plan.
- Full visibility to progress against plan allows supply chain planners to generates feasible, integrated, sales and operations plans based on actual plant capacity and material availability, to help ensure on time, in full delivery at the lowest cost.

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Proven Business Outcomes

Advanced Planning and Scheduling can reduce costs by 20% and improve on time delivery by 30%*

Improved competitive advantage

Reduced Customer Lead Times and Improve On-Time Delivery by up to 30%

Streamlined internal processes

Cut planning time by more than 50%
Save supervisor distractions and align sales, planning, and operations

Improved margins and cash flow

Reduced stock inventory, improved plant throughput and reduced labor costs by up to 20%

Fast ROI

First year ROI through improved schedule efficiency

*For customer case story details please see https://www.planettogether.com/planettogether-advance-planning-scheduling-case-studies

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Advanced Planning and Scheduling Features & Functions

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Advanced Planning & Scheduling ...more than a Scheduling tool

A Realistic Capacity & Material Constraint Model

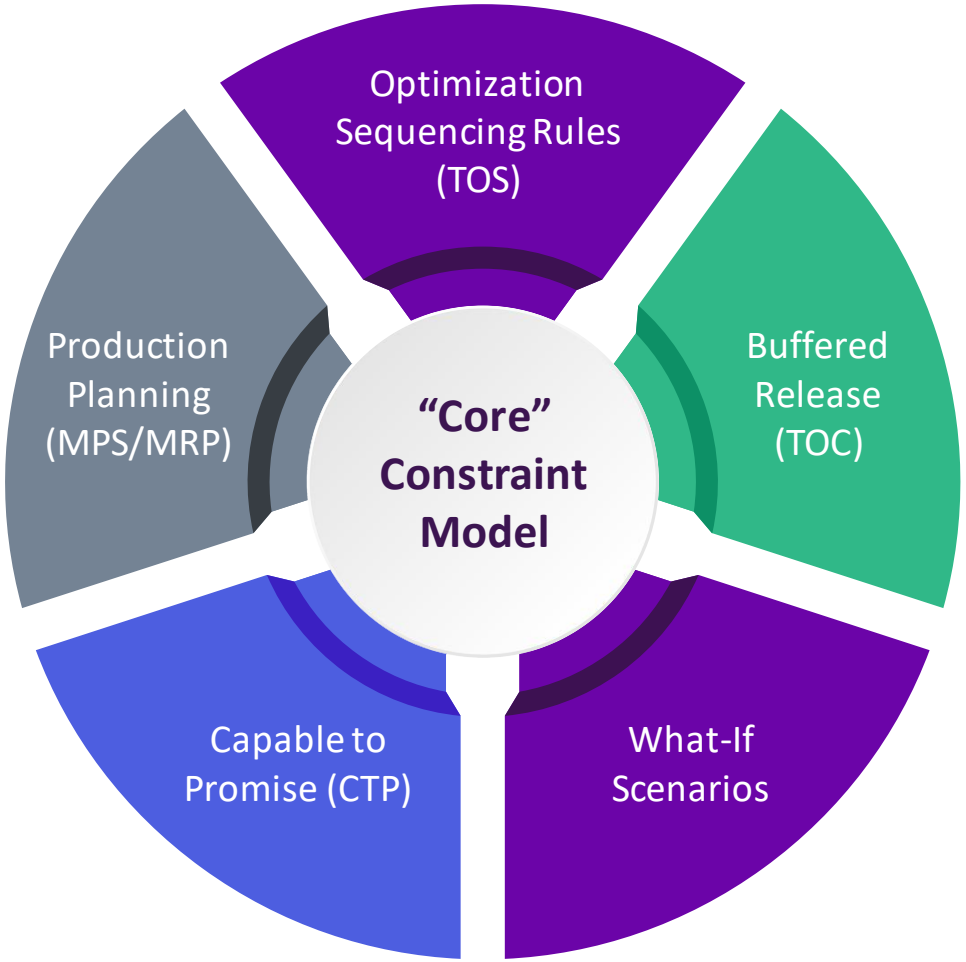
Since 2004 PlanetTogether has developing new ways and refining software capabilities to model manufacturing facilities down to the most finite detail.

Resources, Capabilities, Attributes, Tanks, Labor, Queue Times, Transfer Times, Purchases, BOMs, Routes, Successor Relationships, Overlapping… to name a few…

Why is this important?

Advanced Planning and Scheduling software strength is its ability to provide tactical and strategic insights into your production and to generate accurate schedules.

A realistic model is the foundation for an accurate production schedule, otherwise everything you're trying to accomplish is based on false information.



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Visual Planning & Scheduling

Ultra-flexible Gantt Display

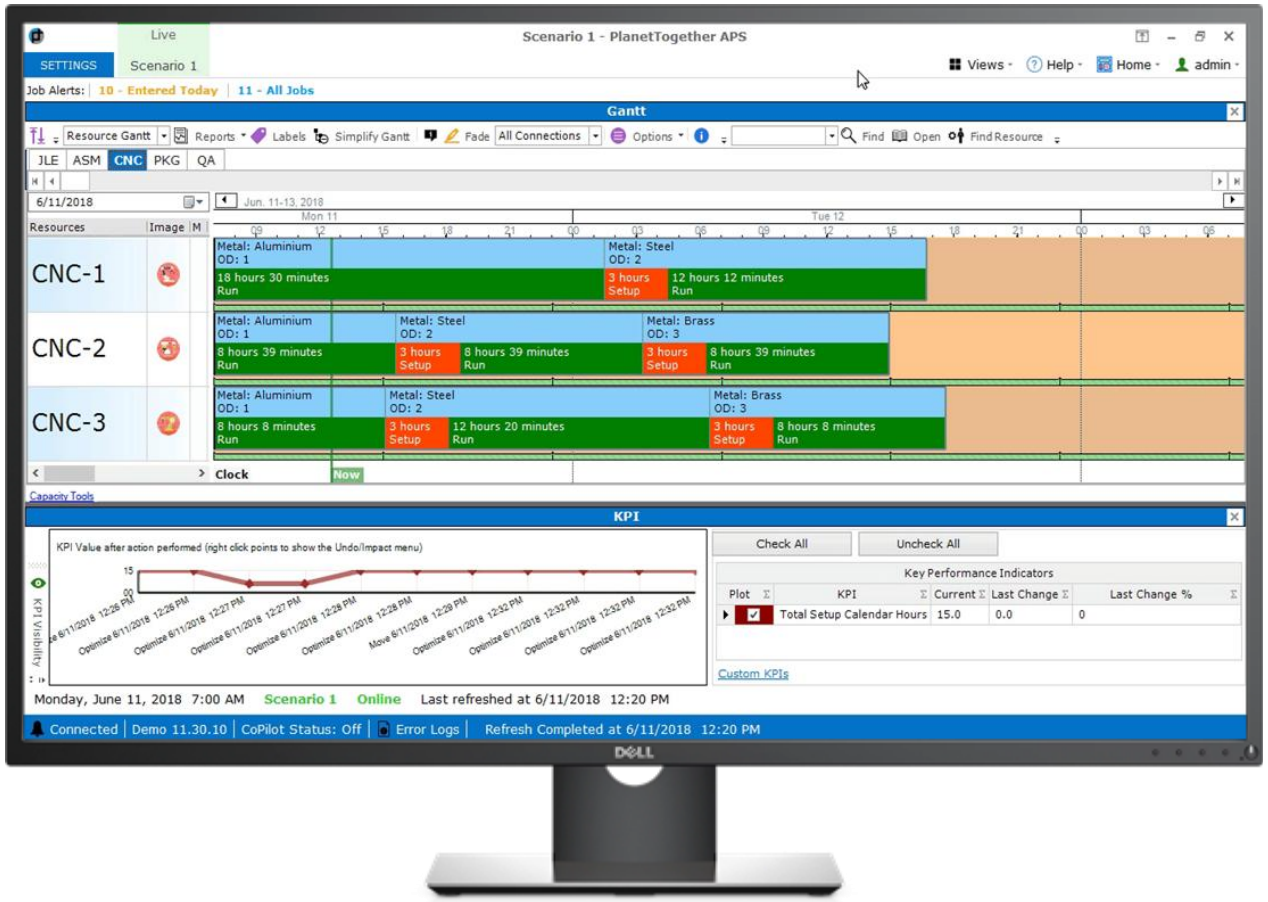
Enterprise wide visibility has never been this clear! See everything you need on one configurable screen. You're in complete control.

Auto-adjusting Drag-&-Drop Constraint-based Scheduling

You'll feel confident knowing your schedules are 100% realistic. No more push back from the shop floor because of unrealistic schedules.

Job & Process Dispatch Lists

Easily publish schedules to the floor on demand or in real-time. One version of reality for the entire company.



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Best-in-class Schedule Optimization (Work Sequencing)

A GPS for your Factory

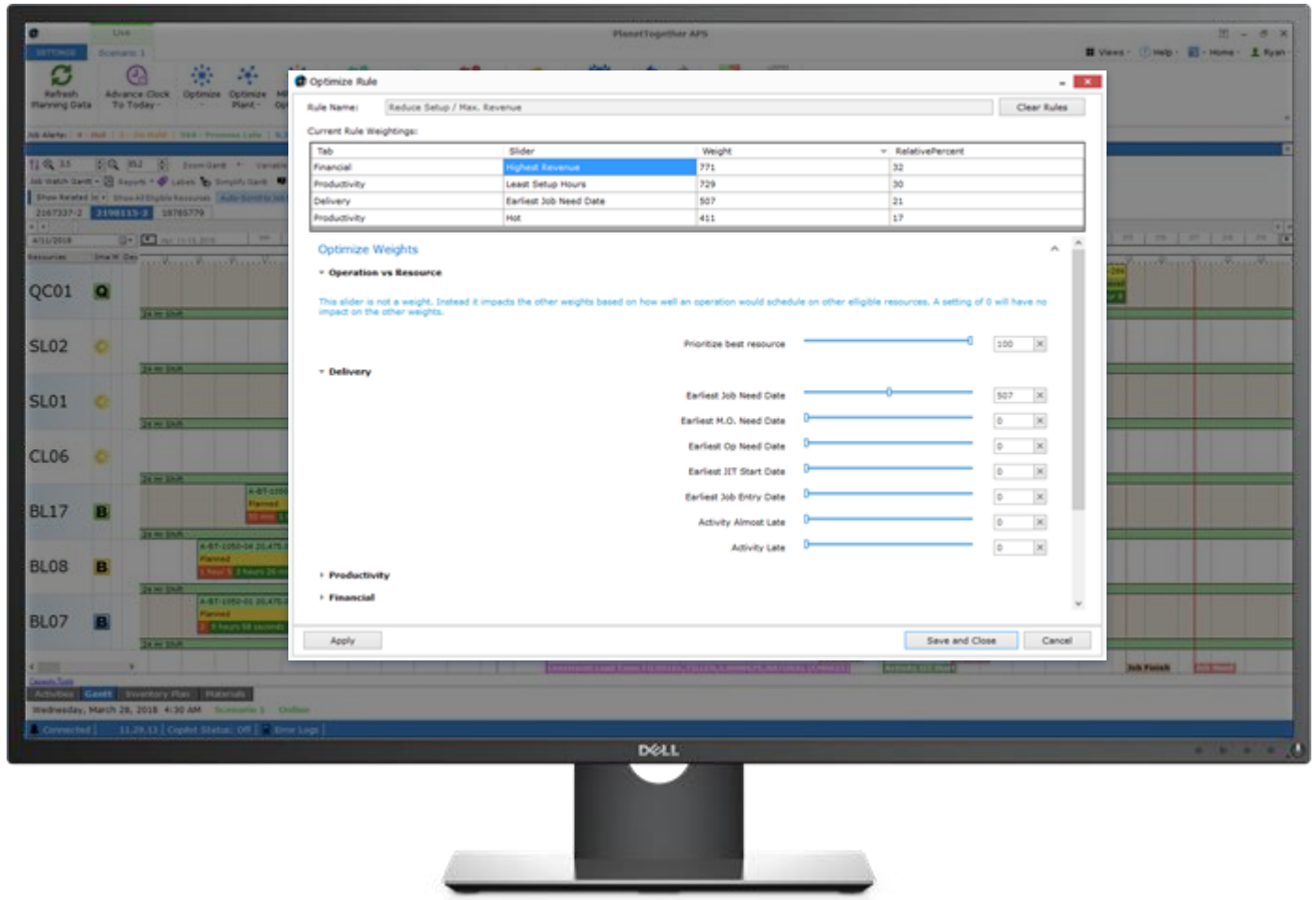
Just like your GPS guides you around traffic, Advanced Planning and Scheduling keeps you on the optimized path as real life events happen. Machine down? New order drop-in? Material shortage? Rest assured you always have the best plan for the situation.

Buffered Release Control

Apply Theory of Constraints (TOC) methodologies and take control over procurement and production variability to maximize throughput and on-time delivery.

Flexible Optimization Rules

As business conditions and goals change, easily change your optimization formula. Simply adjust the sliders to fine tune your sequencing.



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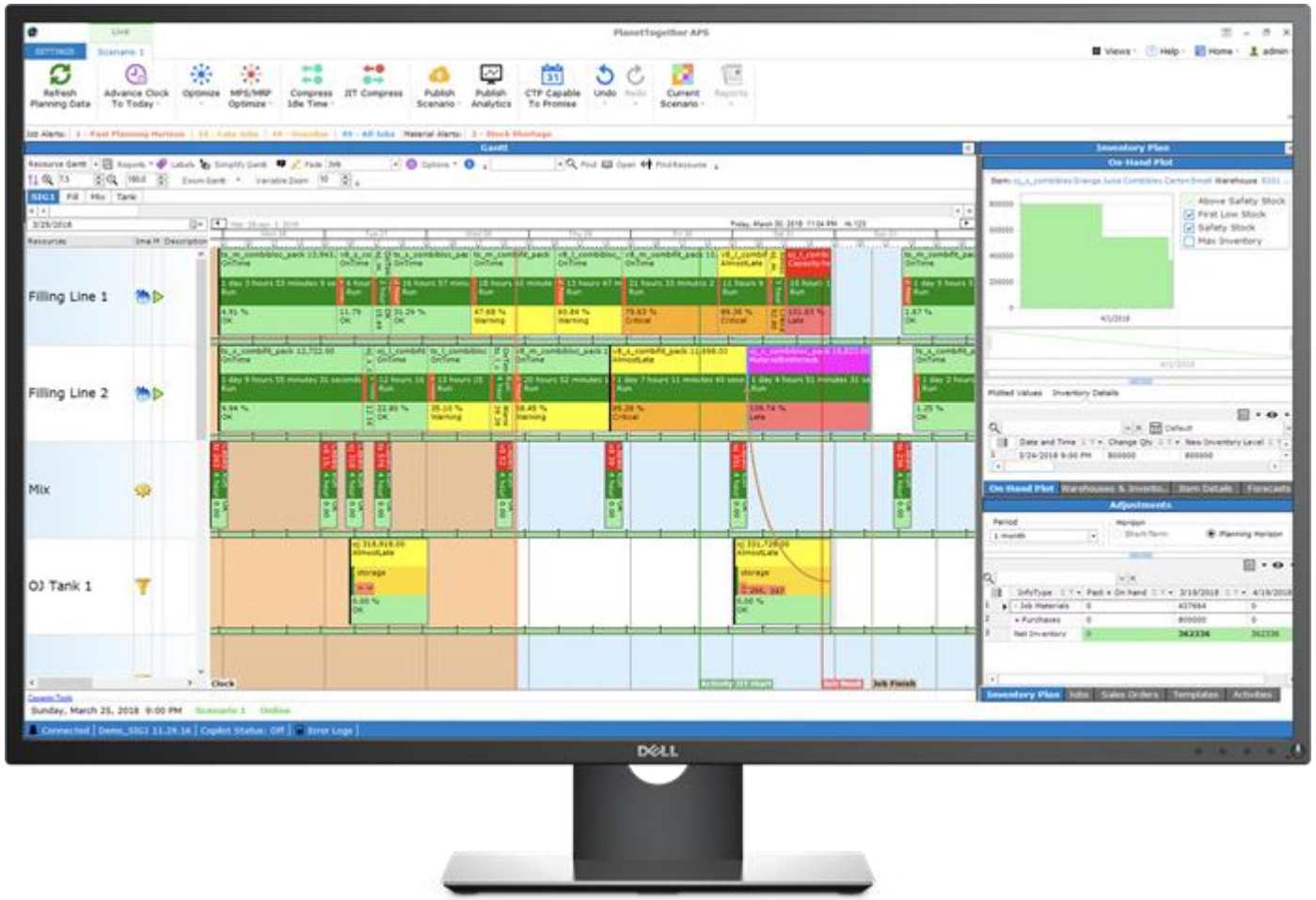
Bottleneck Management



Automatic Bottleneck Avoidance

The Scheduling functionality will automatically shift work to the most readily capable and available resource. That translates to faster flow and lower inventories.

Bottleneck Identification

Easily spot and drill into unavoidable bottlenecks. With visual scheduling and dynamic Gantt labeling it will be easy to tell which step in your process is causing the delay, the first step to resolution

Bottleneck Resolution Tools

Using included tools like Buffer Management Workbench or Overtime Intervals, you'll be able to anticipate future constraints and make the changes needed before it's too late.

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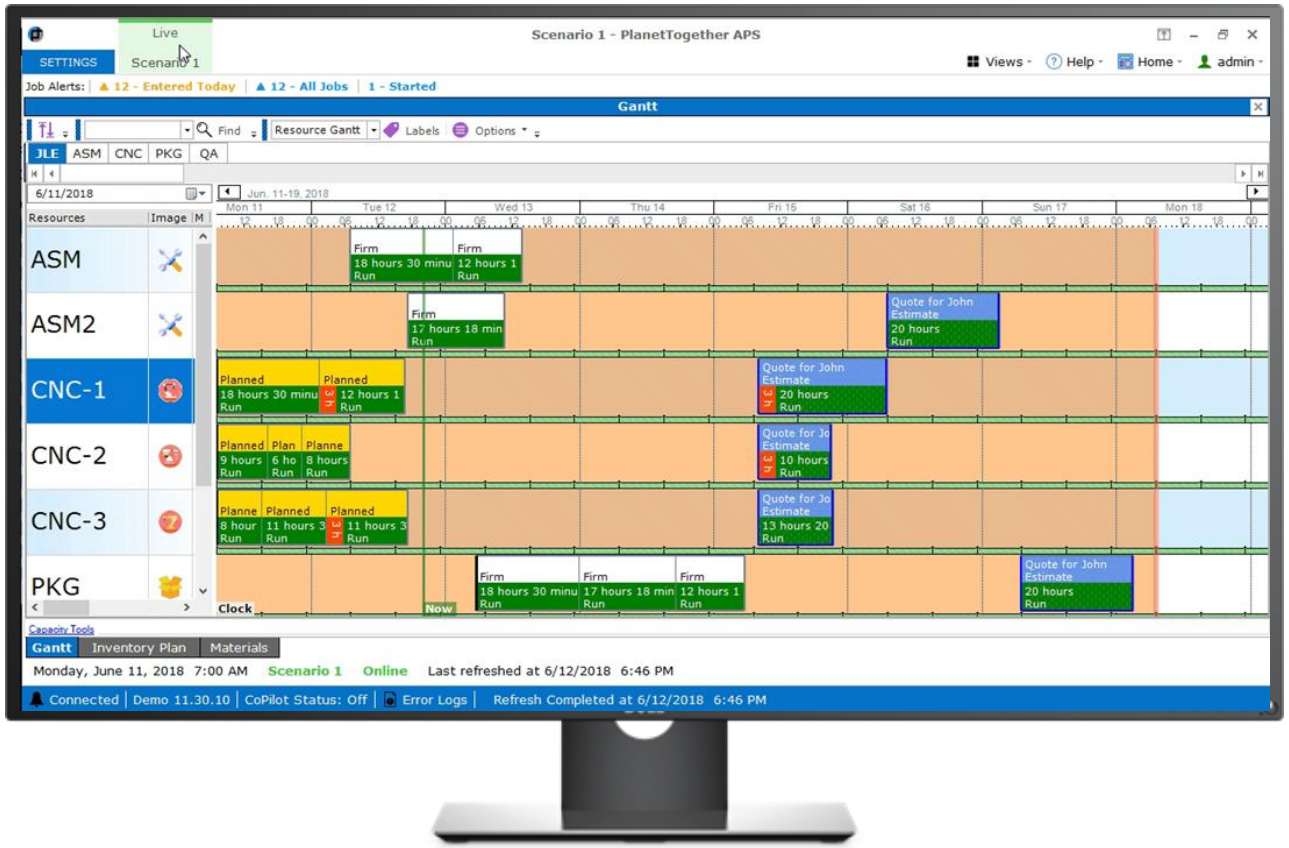
What-if Scenarios ?



Unlimited Scenarios

Included with the Advanced Planning and Scheduling software you get an unlimited number of planning scenarios. Add capacity, load up the forecasts for next year and see the result. All without affecting your live planning and scheduling scenario or master data.

Compare your results

Which scenario is best? Compare your KPIs and find out.

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Business-Driving KPIs

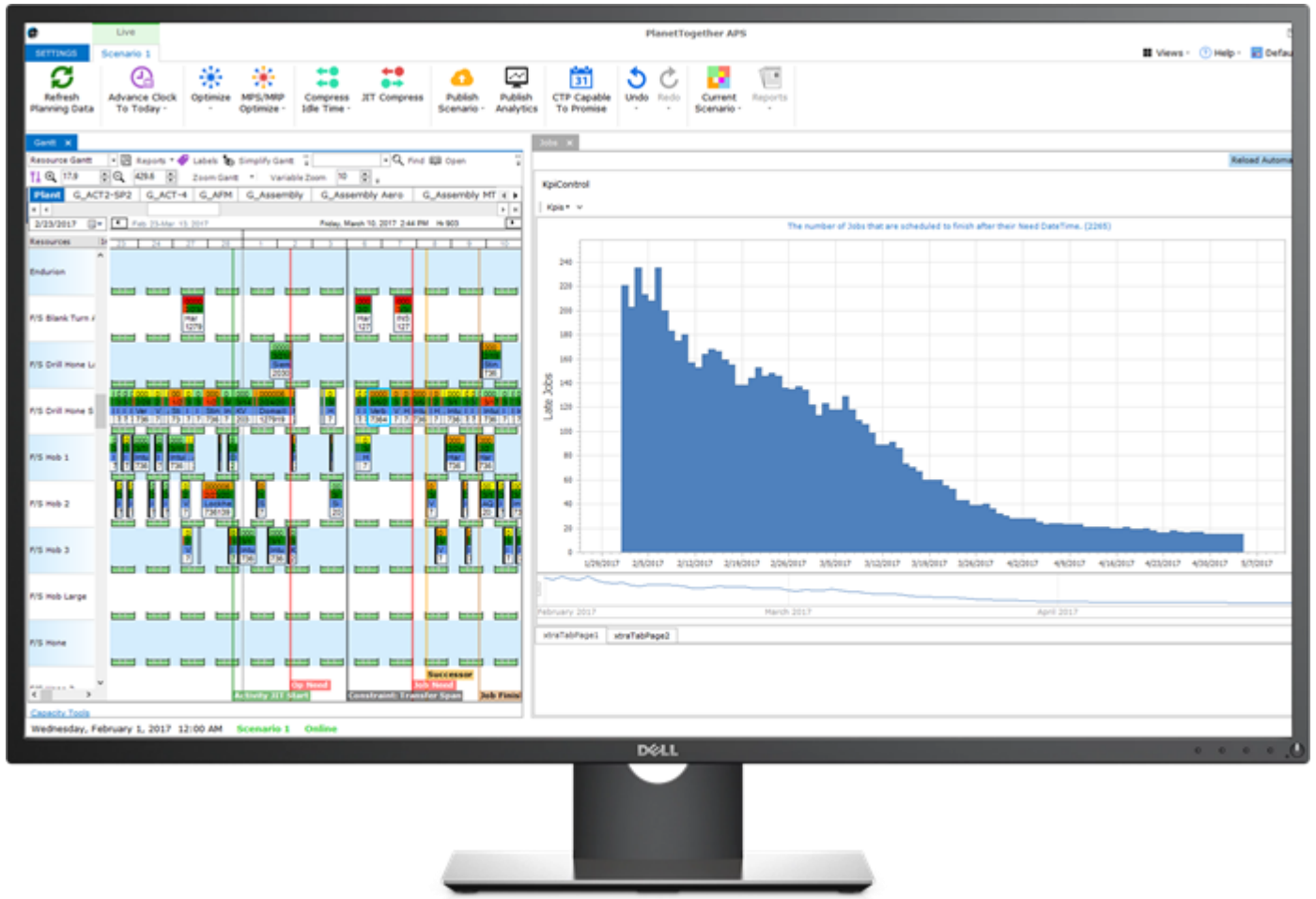
Real-time Impact Analysis

Tired of guessing the impact of changing or expediting orders? Planet Together will instantly show how proposed scheduling changes affect customer orders and forecasts.

Customizable KPIs

How can you improve your plans if you can't measure them? Schedule against business driven metrics and achieve your performance goals.

Share realistic, up-to-date plans and scenarios with your team so everyone knows what to expect.



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Multi-Plant Production Planning

Rapid Demand Planning

Whether it's sales orders, forecasts, estimates or safety stock, align your supply to customer demand by planning the right production jobs and purchase orders -- in seconds.

Alleviate Shortages

With Planning alerts you know exactly when there is a material issue and what's impacted. Quickly spot, drill into the data and resolve the issue weeks or months ahead of impacting customers.

Capable-to-Promise (CTP)

Want to give your customers faster delivery quotes, based on real life capacity, on-hand quantity and incoming material purchases? With the capable to promise planning capabilities you can provide an accurate quote in seconds.

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Integrated Planning and Execution Demo

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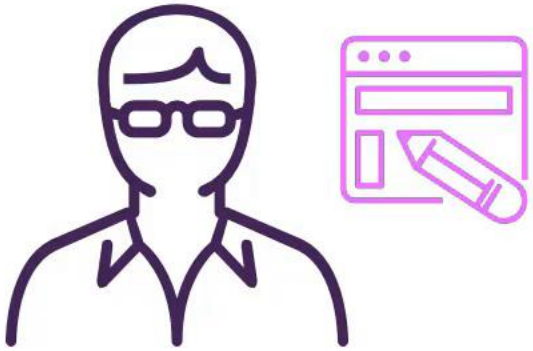
Integrated Planning and Execution Demo Video

Improve Agility & Resilience



Planner

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Customer References

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Advanced Planning and Scheduling References*

From $20M to Fortune 500

Food, CPG, Pharma

Manufacturing

Discrete

*For customer case story details please see https://www.planettogether.com/planettogether-advance-planning-scheduling-case-studies

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Advanced Planning and Scheduling Customer Testimonials

"Our scheduling system was limiting our production capabilities. It had become the determinant of growth potential, not capacity. Scheduling now takes about 15 minutes.

Dick Marx
Materials Manager

KNAPHEIDE

"The requests for information and meetings around challenges tied to production have gone down. We now have 'one version of the truth,' which is a win."

Brian Goodwin
Master Scheduler

"PlanetTogether's software is by far the best and most encompassing planning software package offered today. Before it took hours to schedule our production because scheduling was based on manual reports. Now it only takes seconds!"

Ed Czysz
Production Scheduling Manager

ILLES SEASONINGS & FLAVORS

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AVEVA

Graphic Packaging International

One of the world's largest packaging manufacturers

Challenges

- Lack of Changeover Visibility
- Scheduling Errors
- Reliance on Tribal Knowledge
- Lack of forward-looking capacity utilization to be able to commit capacity to customers

Solution

- Advanced Planning and Scheduling (APS) software from Planet Together

Results

- 6% reduction in unit costs due to increased Overall Equipment Effectiveness (OEE)
- $16M in annual saving
- Scheduling decisions are based on automation and company KPIs rather than intuition

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ABOUT AVEVA

AVEVA is a global leader in engineering and industrial software driving digital transformation across the entire asset and operational life cycle of capital-intensive industries.

The company's engineering, planning and operations, asset performance, and monitoring and control solutions deliver proven results to over 16,000 customers across the globe. Its customers are supported by the largest industrial software ecosystem, including 4,200 partners and 5,700 certified developers. AVEVA is headquartered in Cambridge, UK, with over 4,400 employees at 80 locations in over 40 countries.

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